

Chapter 3. Research Design (Sample)

Working draft · Fictional brand “Shanyueshai” tea-chain study · For CoEditor demo only

3.1 Data and Sample

We assemble a panel of 3,842 store-quarter observations covering 1,208 outlets of the fictitious tea chain “Shanyueshai” between 2024Q1 and 2025Q4. Records come from the internal POS system, cross-validated against settlement reports. We drop outlets operating fewer than three months, outlets closed for more than thirty days within a quarter, observations with missing core variables, and franchise stores with incomplete data feeds.

3.2 Variables

The dependent variable is floor-area-adjusted quarterly revenue per store. Digital intensity is constructed from operation-log mining, counting the share of digital-tool events among all logged operations (four categories, 92 dictionary entries; a one-zero taxonomy). Controls include store size, staffing, membership penetration, outlet age, business-zone type, city tier, and franchise ownership.

3.3 Identification

We estimate a two-way fixed-effects model with store and quarter effects, clustering standard errors at the store level. Endogeneity is addressed through lagged regressors, peer-zone mean digital intensity as an instrument, and nearest-neighbor matching on the digital-intensity median.

This is a synthetic sample document for the CoEditor demo vault. All numbers, brands, and findings are fictitious.